

Sip Your Emotions: Blending Emotion and Data in Cocktail Design

Chenpu Li
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2251319@tongji.edu.cn

Yangfan Tian
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2252037@tongji.edu.cn

Tian Qi
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2254323@tongji.edu.cn

YaQi Yang
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2253848@tongji.edu.cn

Jiajie Chen
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2250232@tongji.edu.cn

Yiting Liu
Tongji University
Tongji University
Shanghai, China
Tongji University
Tongji University
Shanghai, China
2250227@tongji.edu.cn

Guanhong Liu
College of Design and Innovation,
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
College of Design and Innovation,
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
liugh23@tongji.edu.cn

Qing Chen
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
qingchen@tongji.edu.cn

Yang Shi
Intelligent Big Data Visualization Lab
College of Design and Innovation,
Tongji University
Shanghai, China
Intelligent Big Data Visualization Lab
College of Design and Innovation,
Tongji University
Shanghai, China
shiyang1230@gmail.com

Nan Cao
Intelligent Big Data Visualization Lab
Tongji College of Design and
Innovation
Shanghai, China
Intelligent Big Data Visualization Lab
Tongji College of Design and
Innovation
Shanghai, China
nan.cao@gmail.com

Abstract

Emotional embodiment has become an emerging focus in both art therapy and interactive design. This paper presents a data-driven system that transforms users' emotional expressions into tangible cocktail recipes, providing a novel way to externalize and "taste" one's feelings. Through natural language processing and an emotion-embedding approach, we map textual input onto base spirits, flavors, and garnishes. Users can refine their "emotional curve" to produce unique personalized cocktails, each reflecting a distinct affective profile. The system employs a multi-dimensional

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721336>

optimization framework and a visual “recipe sphere” interface, balancing classic mixology principles with individual emotional needs. By uniting artificial intelligence, flavor theory, and an interactive design, this work offers a fresh avenue for emotional embodiment in art therapy, where users gain deeper insight into their own affective states through the creative act of cocktail making.

CCS Concepts

• **Human-centered computing** → **User interface management systems**; *Information visualization*; • **Computing methodologies** → *Lexical semantics*; • **Applied computing** → **Media arts**.

Keywords

Art Therapy, Emotional Embodiment, Cocktail Design, Data Visualization

ACM Reference Format:

Chenpu Li, Yangfan Tian, Tian Qi, YaQi Yang, Jiajie Chen, Yiting Liu, Guan-hong Liu, Qing Chen, Yang Shi, and Nan Cao. 2025. Sip Your Emotions: Blending Emotion and Data in Cocktail Design. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3706599.3721336>

1 Introduction to Emotion-Driven Cocktail Design

Emotions play a significant role in human experience but often remain abstract and difficult to articulate. This project aims to materialize emotions into tangible, sensory creations: personalized cocktails. By combining computational design and mixology, we provide users with a novel way to express, explore, and experience their feelings.

The key innovation lies in the integration of emotion analysis with cocktail design. Unlike traditional AI systems that optimize functionality, our approach emphasizes *emotional embodiment*, allowing users to “sip” their emotions.

2 Mapping Emotions to Flavors

We utilize GPT’s pretrained embeddings to establish semantic relationships between emotional expressions (e.g., “warmth,” “melancholy”) and flavor descriptors (e.g., “citrus,” “floral”). These embeddings form the basis of a mapping algorithm that translates user input into ingredient combinations.

Emotion recognition systems have made significant strides in recent years, particularly in the context of brainwave data. Research by Zhang et al. (2022) and Li et al. (2022) demonstrated the potential of using electroencephalogram (EEG) signals for emotion detection, which has direct implications for improving personalized experiences [2, 8]. In our work, we leverage similar approaches to map emotional states onto flavor profiles, ensuring both emotional alignment and flavor coherence.

To ensure both emotional alignment and flavor coherence, the system employs a multi-dimensional optimization process. This process integrates classic cocktail recipes, incorporates a “taboo table” to avoid incompatible combinations, and applies a “reward table” to prioritize historically proven or particularly harmonious pairings. The application of multi-dimensional optimization in cocktail recipes has been demonstrated in previous work by Viswanath

and Lee (2018) [7], who showed how emotion-driven algorithms can enhance affective computing systems.

By balancing these factors, the system translates the user’s emotional state into a tangible, flavorful cocktail recipe, embodying their feelings through carefully curated ingredients and taste profiles.

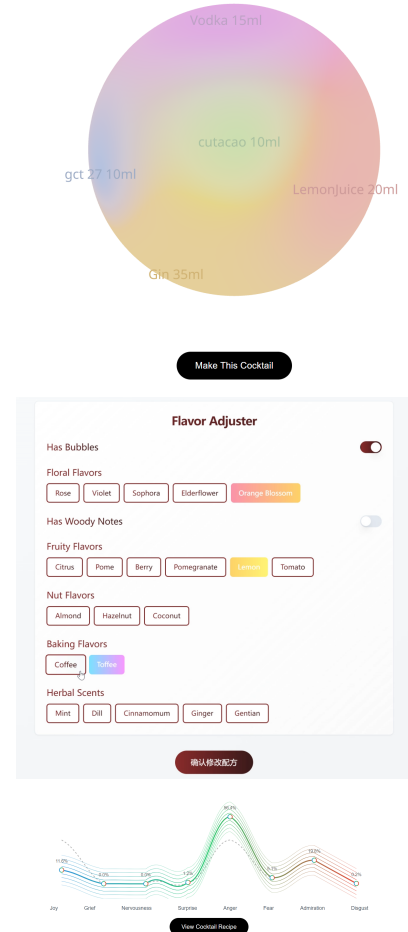


Figure 1: Snapshots of the interactive interface showing real-time updates to cocktail recipes based on emotional inputs.

3 Interactive Design and Applications

The system features an intuitive interface where users can input emotional descriptions, visualize emotional vectors, and fine-tune their “emotional curve.” As shown in Figure 1, the interface dynamically adjusts ingredient ratios and displays the resulting recipe as a 3D “recipe sphere.”

The concept of emotional embodiment, where emotions are translated into physical or sensory forms, has been explored in various contexts. Ekman (1992) and Plutchik (2001) argued that emotions can be categorized into fundamental states, each with specific physical manifestations [1, 4]. Our work builds on this by allowing users

to experience their emotions through the tangible act of cocktail creation. The ability to visualize and modify the emotional profile directly in the interface adds a new layer of interaction to emotional embodiment.

Potential applications include:

- **Art Therapy:** Helping users externalize emotions through creative and sensory experiences. This approach is aligned with the work of Rodrigues (2016), who explored the therapeutic potential of EEG-based emotion detection in art therapy [5].
- **Entertainment:** Enhancing social interactions at parties with custom cocktails, where the cocktail recipes can reflect participants' current emotional states, similar to emotion-driven experiences in virtual environments (Smith et al., 2019) [6].
- **Wellness:** Encouraging mindful reflection and self-expression in a playful context, an idea also supported by Lupu (2019) in her work on multisensory experiences [3].

4 Video Demonstration and Future Directions

Our 5-minute video illustrates the system's workflow, from emotion analysis to recipe generation and physical cocktail preparation. The narrative emphasizes user interaction and the emotional embodiment concept.

Future work will explore expanding the flavor database, incorporating user feedback to refine taste mapping, and integrating

cross-cultural perspectives on emotional expression. As Ekman (1992) and Plutchik (2001) suggested, emotions vary widely across cultures, and it will be essential to account for these differences when further developing the system [1, 4].

References

- [1] P. Ekman. 1992. Are there basic emotions? *Psychological Review* 99, 3 (1992), 550–553.
- [2] J. Li, X. Wang, and Z. Zhao. 2022. Emotion recognition via electroencephalogram (EEG) signal using deep learning techniques. *IEEE Transactions on Affective Computing* 13, 4 (2022), 9625702. <https://ieeexplore.ieee.org/document/9625702>
- [3] C. Lupu. 2019. Wine tourism: A multisensory experience. https://www.researchgate.net/profile/Cristina-Lupu-5/publication/335082608_Wine_tourism_a_multisensory_experience/links/5d4daa88299bf1995b735822/Wine-tourism-a-multisensory-experience.pdf
- [4] R. Plutchik. 2001. The nature of emotions: Fundamental questions. *American Psychologist* 56, 3 (2001), 197–206.
- [5] A. M. T. Rodrigues. 2016. Human Emotion Detection via Brain Waves: Study by Using Electroencephalogram (EEG). https://www.researchgate.net/publication/312422191_Human_Emotion_Detection_via_Brain_Waves_Study_by_Using_Electroencephalogram_EEG
- [6] D. Smith, S. Lee, and K. Kim. 2019. Emotion recognition from brain waves for personalized experience in virtual environments. *Computers in Human Behavior* 98 (May 2019), 159–166. <https://www.sciencedirect.com/science/article/abs/pii/S0950329319309255>
- [7] F. T. Viswanath and M. H. Lee. 2018. EEG based emotion detection for affective computing. *Journal of Neuroscience Methods* 289 (Dec 2018), 45–52. <https://www.sciencedirect.com/science/article/abs/pii/S0924224418307659>
- [8] G. Zhang, Y. Wang, and Y. Chen. 2022. Emotion detection from brain waves for user-centric interface. *IEEE Access* 10 (2022), 10597545. <https://ieeexplore.ieee.org/document/10597545>